Closed Richelot fibres and the actual Ψ kernel

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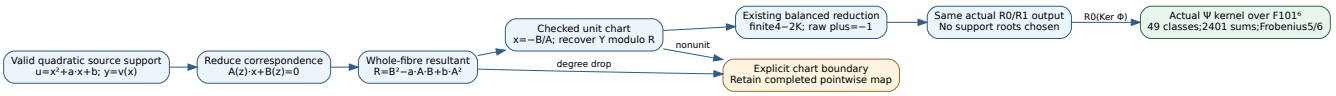
The new evaluator recovers actual Richelot images without splitting quadratic source support. Its targeted comparison passes192 inputs in each R0/R1 direction. The separate kernel test computes49 actual T0 classes R0(Ker Φ), checks every2401 addition,7-annihilation, geometric dualR0 norm2 and Frobenius eigenvalues5/6 over the existing $F101^6$ field. PCCdb5cea/b0738e; the all-domain comparison remains pending here. [pcc:unverified]



Exact finite-algebra elimination

For valid $u=x^2+a\cdot x+b$, reduce the existing correspondence $g1(x)h1(z)+g2(x)h2(z)$ to $A(z)\cdot x+B(z)$. Put $R=B^2-a\cdot A\cdot B+b\cdot A^2$. The identity

$$(A\cdot x+B)(B-a\cdot A-A\cdot x)+A^2u(x)=R$$

shows R vanishes in the fibre algebra. If R has degree4 and A is a unit moduloR, $x=-B/A$ gives mutually inverse algebra maps between $K[z,x]/(u,A\cdot x+B)$ and $K[z]/R$. This preserves nilpotents and multiplicities. For $u=(x-t)^2$ the resultant is $(A\cdot t+B)^2$. No finite-tests-to-global inference is needed for this identity. [pcc:unverified]

If source_scale·target_scale·v(−B/A) is also a unit, recover the target ordinate by the original Richelot equation. Runtime checks source u, the reduced correspondence and the target curve moduloR. The resulting four-point divisor minus2K has raw balanced plus=−1 and uses the unchanged reduction previously used by Cantor addition. Lower-degree supports and infinities reuse the existing completed fibres. [pcc:unverified]

Degree drops and nonunits reject the chart explicitly. Prime inputs retain the old completed map as fallback; extension inputs report a boundary. This does not assert that an unsupported point or map does not exist. [pcc:unverified]

Kernel, proof scope and continuing checks

Actual49-point geometric kernel

The retained global square is Ψ·R0=R1·Φ, with degΨ=49. R0 has2-primary kernel, hence is injective on the49 geometric7-torsion points KerΦ. Their images are49 distinct points killed byΨ and therefore its full geometric kernel. The new chart computes the actual balanced T0 divisor classes in the existing $F101^6$ field. [pcc:unverified]

Native02 checks every image,7-annihilation, all49 geometric dualR0 compositions against[2], all2401 additions and Frobenius5/6. The independent flat-field elliptic route supplies actual7-kernels and checks Φ killing. PCCb0738e pins a partial process receipt that already contains this test's PASS/ok; its separate full-domain test is not thereby a pass. [pcc:unverified]

Faithful partial Lean proof

Six ClosedFibres.lean identities pass gate and all local declaration bindings, rootPCCa97f41: quadratic remainder, resultant norm identity, linear/quadratic source recovery under explicit inverse, repeated-support resultant and cleared ordinate recovery. Total source census103, zero trusted native. Full coordinate-ring, scheme/Picard and projective geometry remain the written bridges in CLOSED_FIBRES.md/rootproof0e5bbf. [pcc:unverified]

Primary source: Kunzweiler, DCC92(2024), §3.2 Propositions7–8. Its later degree5/rational-Weierstrass assumptions are not imported as coverage for our split sextics. Existing native/consumer/generic Richelot and balanced engines are reused. KB retrieval is unavailable; no novelty is claimed. [pcc:unverified]

Current scope and next action

The complete four-domain12317-class comparison is still running. Require old/closed equality, both dual2 identities, bijections, paired additions, coefficient descent and wrong-model rejection. Exercise actual geometric chart boundaries, retain the old evaluator, then integrate and run the complete changed guard and109-source/Psi controls before publication. [pcc:unverified]

Read CLOSED_FIBRES_STATUS.md, src/closed_fibres.rs and immutable evidence with the closed-fibres prefix. This branch starts readyPR1490/994869034. Parent CI and both frozen-ledger forks remain separate obligations. Next CT24 retains the further nondual7-hop. S3 admission/livebudget block benchmarks and paid work. The current result concerns explicit evaluation and geometric kernel data; development elapsed times do not measure cost. [pcc:unverified]